DISCRETE STRUCTURES

CMSC250

Chapter 2: Introduction to Proposition Logic

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Compound statements, translations, truth tables, and precise reasoning



Notation. We write $\sim p$ for "not p." The symbol $\neg p$ is also common in mathematics and means exactly the same thing. We will use $\sim$ consistently in these notes.

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1 Why propositional logic needs a formal language

Everyday language can hide important differences. Consider "I will submit the project or attend office hours." Does the sentence allow both activities? In ordinary conversation, context may decide. In mathematical logic, the symbol $\vee$ has a precise meaning: at least one of the statements must be true, and both are allowed.

The goal of propositional logic is to replace ambiguity with a language whose meaning can be calculated. We start with atomic propositions such as p and q, combine them using connectives, and evaluate the resulting expression for every assignment of truth values. This allows us to reason about a compound statement without knowing the particular real-world facts represented by p and q.

2 Atomic and compound propositions

An **atomic proposition** is a statement treated as one indivisible logical unit. For example, let p mean "The assignment is complete" and q mean "The assignment was submitted." We do not inspect the words inside p or q when building a truth table: each simply receives either 1 (true) or 0 (false).

A **compound proposition** combines simpler propositions using connectives. For example, $p \wedge q$ says the assignment is complete *and* submitted; $p \vee q$ says at least one of those conditions holds; $\sim p$ says the assignment is not complete. Parentheses identify which parts are combined first.

2.1 The three basic connectives

Symbol	Name	Meaning
$\sim p$	Negation	p is false.
$p \wedge q$	Conjunction	Both p and q are true.
$p \vee q$	Disjunction	At least one of p, q is true (inclusive OR).

p	q	$\sim p$	$p \wedge q$	$p \vee q$
1	1	0	1	1
1	0	0	0	1
0	1	1	0	1
0	0	1	0	0

Key idea. Negation reverses one truth value; AND requires every component to be true; inclusive OR requires at least one component to be true.

3 Parsing expressions and the role of parentheses

An expression such as $\sim (p \wedge q)$ means "it is not true that both p and q hold." The expression $(\sim p) \wedge q$ means "p is false and q is true." These are different claims. In the first, the case $p = 0, q = 0$ makes the expression true; in the second, it makes the expression false.

p	q	$\sim (p \wedge q)$	$(\sim p) \wedge q$
1	1	0	0
1	0	1	0
0	1	1	1
0	0	1	0

3.1 Order of operations

Unless parentheses indicate otherwise, the usual precedence among the three connectives is negation first, conjunction second, and disjunction third:

$$\sim \quad \text{before} \quad \wedge \quad \text{before} \quad \vee .$$

Thus $\sim p \vee q \wedge r$ is read as $(\sim p) \vee (q \wedge r)$. For longer expressions, however, explicit parentheses make the intended structure easier to verify and are strongly recommended in written proofs and translations.

Worked example: evaluate a compound statement

Evaluate $\sim (p \vee q) \wedge r$ when $p = 0, q = 0, r = 1$. First, $p \vee q = 0$. Negating gives $\sim (p \vee q) = 1$. Finally, $1 \wedge r = 1 \wedge 1 = 1$. The result is true.

If instead $p = 1, q = 0, r = 1$, then $p \vee q = 1$, its negation is 0, and the entire conjunction is 0. The value of r cannot rescue a conjunction whose other part is false.

4 Translating English carefully

Translation is easiest when we first identify the atomic statements and then ask which combinations of their truth values satisfy the English sentence. Consider $L =$ "I wear a sock on my left foot" and $R =$ "I wear a sock on my right foot," assuming at most one sock per foot.

English	Logical expression
Both feet have socks.	$L \wedge R$
At least one foot has a sock.	$L \vee R$
Neither foot has a sock.	$\sim L \wedge \sim R$
Exactly one foot has a sock.	$(L \wedge \sim R) \vee (\sim L \wedge R)$
At most one foot has a sock.	$\sim (L \wedge R)$
Only the left foot has a sock.	$L \wedge \sim R$
It is false that neither foot has a sock.	$\sim (\sim L \wedge \sim R)$

4.1 At least, at most, and exactly

These phrases refer to *counts of true statements*. For two propositions, "at least one" permits one or two true statements; "at most one" permits zero or one; and "exactly one" permits one and forbids both zero and two.

L	R	At least one	At most one	Exactly one
1	1	1	0	0
1	0	1	1	1
0	1	1	1	1
0	0	0	1	0

Key idea. A useful translation check: list the interpretations that should make the English sentence true. Your logical formula must have value 1 on exactly those rows.

4.2 Translating conditions over three intervals

Suppose m, a, n mean "Jack was awake during the morning, afternoon, and night," respectively. We treat each interval as an all-or-nothing unit, as the question's setup intends.

- "Jack was awake in at least one interval": $m \vee a \vee n$.
- "Jack was asleep in all three intervals": $\sim m \wedge \sim a \wedge \sim n$.
- "Jack was asleep in at least one interval": $\sim m \vee \sim a \vee \sim n$.
- "Jack was not awake in all three intervals": $\sim (m \wedge a \wedge n)$.

Notice that the last two sentences describe the same situations. We can verify that claim by examining all eight interpretations, not merely by deciding that the English sounds similar.

Worked example: two people, three time periods

Let m, a, n describe Jack's awake intervals, and o, f, i describe Tony's corresponding intervals. Translate "During each of the three intervals, at least one of the two was awake."

For the morning we need $m \vee o$. For the afternoon we need $a \vee f$. For the night we need $n \vee i$. Since the requirement applies to *every* interval, all three conditions must hold:

$$\boxed{(m \vee o) \wedge (a \vee f) \wedge (n \vee i).}$$

A common mistake is to write $(m \wedge a \wedge n) \vee (o \wedge f \wedge i)$, which says one person was awake for *all* intervals. That is unnecessarily restrictive: the two people could alternate being awake.

5 Constructing complete truth tables

A truth table lists every possible interpretation of the variables. With n independent variables there are 2^n rows. We use **alphabetical variable order**, **1 and 0**, and **descending binary order**. For three variables, the first row is 111 and the last is 000.

p	q	r
1	1	1
1	1	0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1
0	0	0

5.1 Why the row pattern works

Think of the truth values as binary digits. The rightmost variable alternates on every row, the middle variable changes every two rows, and the leftmost variable changes every four rows. Starting at the binary number 111 and counting down to 000 guarantees that every combination appears once and only once.

5.2 Add intermediate columns

When evaluating a nested expression, calculate smaller subexpressions first. Consider

$$F = (p \vee q) \wedge (\sim p \vee \sim q).$$

The first parenthesis requires at least one true variable. The second requires at least one false variable. Together they require *exactly one* of p, q to be true.

p	q	$\sim p$	$\sim q$	$p \vee q$	$\sim p \vee \sim q$	F
1	1	0	0	1	0	0
1	0	0	1	1	1	1
0	1	1	0	1	1	1
0	0	1	1	0	1	0

Worked example: a three-variable truth table

Evaluate $G = r \wedge (\sim p \vee q)$. We first calculate $\sim p$, then $\sim p \vee q$, and finally combine the result with r.

p	q	r	$\sim p$	$\sim p \vee q$	G
1	1	1	0	1	1
1	1	0	0	1	0
1	0	1	0	0	0
1	0	0	0	0	0
0	1	1	1	1	1
0	1	0	1	1	0
0	0	1	1	1	1
0	0	0	1	1	0

The final column has a 1 only in rows where $r = 1$ and either $p = 0$ or $q = 1$. It is useful to describe the meaning of the result in words: this makes it easier to catch an incorrect intermediate column.

6 Logical equivalence from truth tables

Two expressions are **logically equivalent** when they have the same truth value under *every* interpretation. We write $A \equiv B$, not $A = B$; equivalence compares logical behavior across all rows.

Worked example: two different formulas, one meaning

Compare

$$A = (p \vee q) \wedge (\sim p \vee \sim q), \qquad B = (p \wedge \sim q) \vee (q \wedge \sim p).$$

p	q	A	B
1	1	0	0
1	0	1	1
0	1	1	1
0	0	0	0

Because the last two columns match on *all four rows*, $A \equiv B$. Both expressions say "exactly one of p and q is true." This is a proof by complete truth table: there are no interpretations left unchecked.

6.1 How to show expressions are not equivalent

To disprove equivalence, you need only *one* row on which the truth values differ. Compare $p \vee q$ and $p \wedge q$. Taking $p = 1, q = 0$ makes the first expression true and the second false. Therefore they are not equivalent. This row is a **counterexample** to the claim of equivalence.

Key idea. Proving equivalence requires agreement on every row. Disproving equivalence requires only one row of disagreement.

7 Tautologies, contradictions, and contingencies

A **tautology** is true under every interpretation; a **contradiction** is false under every interpretation; a **contingency** is true for some interpretations and false for others.

p	$p \vee \sim p$	$p \wedge \sim p$	p
1	1	0	1
0	1	0	0

The first expression is a tautology: either p is true or it is not. The second is a contradiction: p cannot be simultaneously true and false. The third is a contingency because its value depends on the interpretation.

These classifications are about the *whole expression*, not about whether one particular assignment happens to make it true or false. For example, $p \vee q$ is true when $p = q = 1$, but that does not make it a tautology; it is false when $p = q = 0$.

8 Common mistakes and grading pitfalls

1. **Treating OR as exclusive.** $p \vee q$ is true when both statements are true. Use an explicit exactly-one formula when both must not hold.
2. **Negating the wrong scope.** $\sim (p \vee q)$ differs from $(\sim p) \vee q$. Draw parentheses before evaluating.
3. **Confusing "not both" with "neither."** "Not both" is $\sim (p \wedge q)$; "neither" is $\sim p \wedge \sim q$.
4. **Skipping interpretations.** A truth table with three independent variables must contain eight rows, even if some rows appear uninteresting.
5. **Using the wrong table convention.** Keep variable columns alphabetical, use 1/0, and list rows in descending binary order.
6. **Claiming equivalence after checking only examples.** Matching on one or two rows is not enough. Every interpretation must match.
7. **Using $=$ instead of $\equiv$.** For logical equivalence, the correct notation is $\equiv$.

9 Practice problems

Try these without consulting the solutions. When constructing a table, include useful intermediate columns.

1. Let $p =$ "The lab is open" and $q =$ "The tutor is present." Translate: (a) both; (b) neither; (c) exactly one; (d) at most one.
2. Let a, b, c mean three switches are on. Express: (a) all three are on; (b) at least one is off; (c) exactly one is on; (d) none is on.
3. Evaluate $\sim (p \wedge q) \vee r$ for $p = 1, q = 1, r = 0$, showing intermediate steps.
4. Build a complete truth table for $(p \vee q) \wedge \sim q$.
5. Build a complete truth table for $r \vee (\sim p \wedge q)$, using the required eight-row ordering.
6. Use a truth table to determine whether $\sim (p \vee q)$ and $(\sim p) \wedge (\sim q)$ are equivalent.
7. Give one interpretation proving $p \vee (q \wedge r)$ and $(p \vee q) \wedge r$ are not equivalent.
8. Classify $p \vee \sim p$, $p \wedge \sim p$, and $(p \vee q) \wedge \sim p$ as tautology, contradiction, or contingency.
9. Let m, a, n represent whether a person was awake during three separate intervals. Translate "awake during at most one interval."
10. Explain why $\sim (p \wedge q)$ does *not* mean the same thing as $\sim p \wedge \sim q$. Give a specific interpretation distinguishing them.

10 Detailed solutions

1. (a) $p \wedge q$; (b) $\sim p \wedge \sim q$; (c) $(p \wedge \sim q) \vee (\sim p \wedge q)$; (d) $\sim (p \wedge q)$. The distinction between (c) and (d) is that (d) permits neither condition.

2. (a) $a \wedge b \wedge c$; (b) $\sim a \vee \sim b \vee \sim c$; (c) $(a \wedge \sim b \wedge \sim c) \vee (\sim a \wedge b \wedge \sim c) \vee (\sim a \wedge \sim b \wedge c)$; (d) $\sim a \wedge \sim b \wedge \sim c$.

3. $p \wedge q = 1 \wedge 1 = 1$, so $\sim (p \wedge q) = 0$. Therefore $0 \vee r = 0 \vee 0 = 0$.

4.

p	q	$p \vee q$	$\sim q$	$(p \vee q) \wedge \sim q$
1	1	1	0	0
1	0	1	1	1
0	1	1	0	0
0	0	0	1	0

5.

p	q	r	$\sim p$	$\sim p \wedge q$	$r \vee (\sim p \wedge q)$
1	1	1	0	0	1
1	1	0	0	0	0
1	0	1	0	0	1
1	0	0	0	0	0
0	1	1	1	1	1
0	1	0	1	1	1
0	0	1	1	0	1
0	0	0	1	0	0

6.

p	q	$\sim (p \vee q)$	$\sim p \wedge \sim q$
1	1	0	0
1	0	0	0
0	1	0	0
0	0	1	1

The final columns match in every row, so the expressions are logically equivalent.

7. Choose $p = 1, q = 0, r = 0$. Then $p \vee (q \wedge r) = 1 \vee 0 = 1$, but $(p \vee q) \wedge r = 1 \wedge 0 = 0$. One disagreement is sufficient.

8. Tautology; contradiction; contingency. For the third expression, $p = 0, q = 1$ makes it true, while $p = 1, q = 1$ makes it false.

9. "At most one" means we forbid every pair of intervals from both being on: $\sim (m \wedge a) \wedge \sim (m \wedge n) \wedge \sim (a \wedge n)$. This permits zero or one awake interval and forbids two or three.

10. Choose $p = 0, q = 1$. Then $\sim (p \wedge q) = \sim 0 = 1$, whereas $\sim p \wedge \sim q = 1 \wedge 0 = 0$. Hence the formulas differ.